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SIMATS ENGINEERING
ASSESSMENT TOOLS

1 A PDA is designed to recognize a^nb^n, but it fails for a^nb^nC^n. Critique the design and identify possible flaws in **stack operations and state transitions**. How would you improve it?

2 A student claims that **Context-Free Grammars (CFGs)**

E → E + T | T

T → T * F | F

F → (E) | **id**

can be directly converted into a Deterministic PDA (DPDA).Critically evaluate this statement with justification and suggest corrections if needed

3 A Turing Machine is designed using **multiple tracks** to solve a problem. Suggest alternative **programming techniques** like checking-off symbols or subroutines to do subtraction

4 While designing a PDA accepts the language

$$L = \{w \mid w \in (a+b)* \ and \ n_a(w) == n_b(w)\}$$

$n_a(w)$ is the number of a's in w

$n_b(w)$ is the number of b's in w, acceptance is defined only by **final state**, ignoring empty stack acceptance. Critique this design choice. Would it affect the language recognized? Explain with reasoning.

5 A system designer suggests replacing a Turing Machine with a PDA to reduce computational complexity. Critically compare **FA, PDA, and Turing Machine** capabilities and evaluate whether this replacement is valid.

PDA is designed to recognize $a^n b^n$, but it fails for $a^n b^n c^n$

PDA $L_1 = \{a^n b^n \mid n \geq 0\}$

but it fails to recognize

$L_2 = \{a^n b^n c^n \mid n \geq 0\}$

How does PDA recognize $a^n b^n$?

A PDA has

- input tape
- finite control
- stack

<u>Processing</u>

Input	Stack operation
a	Push A
a	Push A
a	Push A
b	POP A
b	POP A
b	POP A

At the end:

- Input is completely read.
- Stack has returned to its initial condition
- Therefore, the string is accepted.

$a^3 b^3$ Accepted.

Improvement:

A standard PDA cannot recognize

$$L = \{ a^n b^n c^n \mid n \geq 0 \}$$

because this language is not context-free

Conclusion: The failure is not merely a transition error; the PDA model itself is insufficient for this language.

Q. Can the given CGF be directly converted to a DPDA?

$E \to E+T \mid T$

$T \to T*F \mid F$

$F \to (E) \mid id$

The claim is not completely correct

$E \to E+T$

$T \to T*F$

<u>Correction:-</u>

$E \to TE'$

$E' \to +TE' \mid \epsilon$

$T' \to *FT' \mid \varepsilon$

$F \to (E) \mid id$

Conclusion: A CGF cannot always be diretly converted into a DPDA. The grammar must satisfy suitable determinism requirements.

Alternative TM techniques instead of multiple tracks:

A turning machine is designed using multiple tracks to solve a problem. Suggest alternative programming techniques like checking -off symbols or sub rotines to do subtraction.

1) Checking -off symbols:-

- mark symbols that have already been proccessed
- for subtraction, match one symbol from one number with one symbol from the other.
- Continue until the required symbols are exhausted
- The unmatched symbols represent the result.

2. Sub routines:-

- find the beginning of a number
- find an unmatched symbol
- mark a symbol
- move to another number
- return to the starting position.

Conclusion:- checking - off symbols and subroutines can reduce the complexity of transition design while achieving the same computation.

4) PDA accepting by final state:-

$L = \{ w \mid w \in (a+b)^* , n_a(w) = n_b(w) \}$

where $n_a(w)$ is the number of as and $n_b(w)$ is the number of bs.

Accepting by final state is not a problem by itself.

Accepting only by defined by:

for a PDA , acceptance can be defined by:

1) final State

2) Empty Stack.

5.

Feature	FA	PDA	Turing machine.
memory	No stack	one stack	unlimited tape
Power	lowest	more powerful than FA.	most powerful
Recognizes	Regular languages	Context-free languages	Recursively enumerable languages.
ex	$a^* b^*$	$a^n b^n$	$a^n b^n c^n$
Can handle complex Computation	No	limited	Yes.

FA:-

- Has only finite memory
- Suitable for regular languages
- Cannot recognize $a^n b^n$

PDA:

- Has a stack
- Can recognize context-free languages
- can recognize $a^n b^n$
- cannot recoginize $a^n b^n c^n$

Turning machine:-

- Has an unlimited read / write tape
- Can Perform general computations.
- Can recognize languages beyond the Capability of PDA

evalution of replacement:

Replacing a Turning Machine with a PDA is not generally valid.

→ PDA has less computational power than a turning machine. Although PDA may be faster or simpler for some specific problems, it cannot be solve every problem that a turning machine can solve.

Conclusion:-

$$\boxed{FA < PDA < TM}$$

In terms of computational power. Therefore a TM should not be replaced by PDA merely to reduce complexity unless the particular problem is known to be context-free.

SIMATS ENGINEERING
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RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

88/100